

4. 弹性支承超静定结构的计算

5-31 图 5-153 (a) 所示结构, 取图 5-153 (b) 为力法基本体系, 其力法典型方程为 ()。 k 为弹簧刚度。(中南大学 2012)

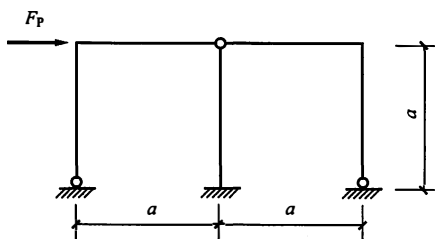


图 5-151

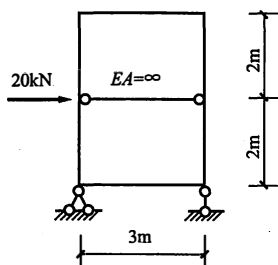


图 5-152

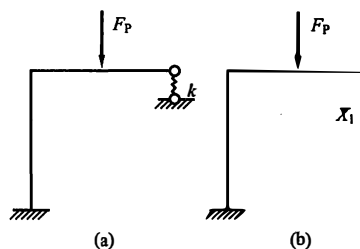


图 5-153

5-32 求图 5-154 所示结构在给定基本体系下的力法方程及方程中的所有系数。(青岛理工大学 2008)

5-33 试计算图 5-155 所示具有弹性支座的结构, 并作弯矩图。抗转刚度系数 $k_\theta = \frac{3EI}{l}$ 。(中国矿业大学 2008)

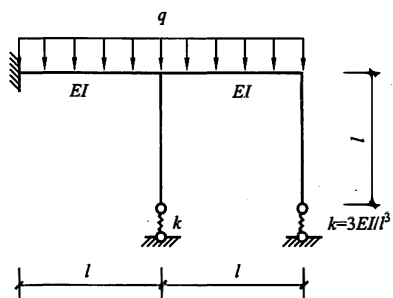


图 5-154

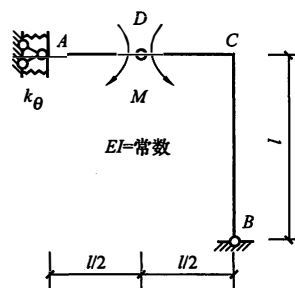
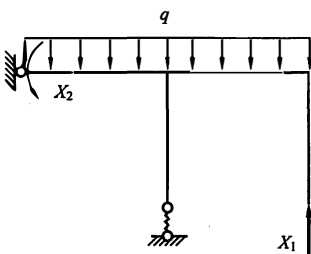


图 5-155

5. 支座发生位移和温度变化时超静定结构的计算

5-34 图 5-156 中等 EI 值单跨梁两侧温度变化时, M 图位于 ()。(青岛理工大学)



学 2008)

A. 低温侧

B. 高温侧

C. 不同侧

D. 零

5-35 判断题。取图 5-157 所示结构基本体系, 则力法方程为 $\begin{cases} \delta_{11}X_1 + \delta_{12}X_2 + \Delta_{1c} = \Delta_1 \\ \delta_{21}X_1 + \delta_{22}X_2 + \Delta_{2c} = \Delta_2 \end{cases}$ 中,

$\begin{cases} \Delta_{1c} = 0 \\ \Delta_1 = -b^\circ \end{cases}$ () (福州大学 2011)

5-36 图 5-158 所示刚架除承受荷载外, 支座 A 下沉了 $\Delta = 2\text{cm}$, 且顺时针转动了 $\theta = 0.01\text{rad}$ 。已知 $E = 2.1 \times 10^4 \text{kN/cm}^2$, $I = 2000 \text{cm}^4$, 用力法求此刚架的弯矩图。(华南理工大学 2009)

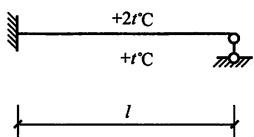


图 5-156

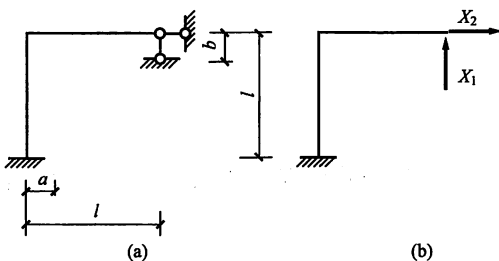


图 5-157

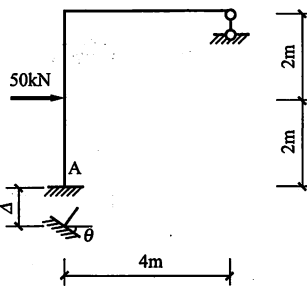


图 5-158

6. 超静定结构的位移计算

5-37 图 5-159 所示梁 (跨长 l) 支座位移产生的跨中点竖向位移为 ()。(哈尔滨工业大学 2011)

A. $l\varphi/16$

B. $l\varphi/8$

C. $l\varphi/4$

D. $l\varphi/2$

5-38 图 5-160 所示为计算得到的超静定刚架的弯矩图, 则 C 点的角位移 $\varphi_C =$ _____, E 点的水平位移 $\Delta_{HE} =$ _____。(湖南大学 2011)

5-39 图 5-161 所示一端固定、另一端定向支承的等截面梁, 当 A 端发生单位转角 $\theta = 1$ 时, B 端的竖向位移 $\Delta =$ _____。(重庆大学 2010)

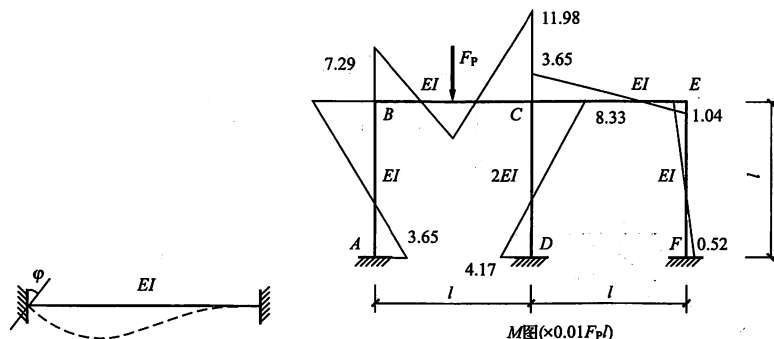


图 5-159

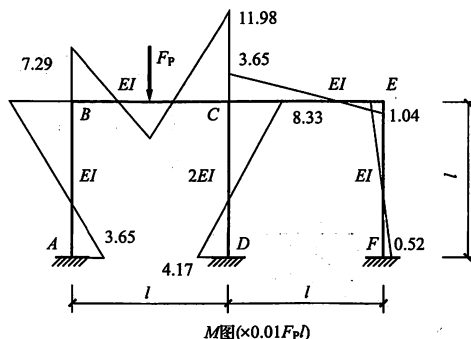


图 5-160

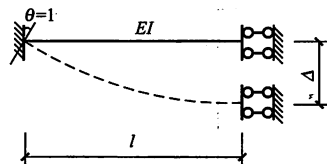


图 5-161

5-40 图 5-162 所示结构, 各杆抗弯刚度 EI 为常数, 忽略杆件的轴向变形。(1) 根



据结构与荷载的特点，选用最简便的方法作刚架的弯矩图；（2）求 C 点的水平位移。（武汉大学 2011）

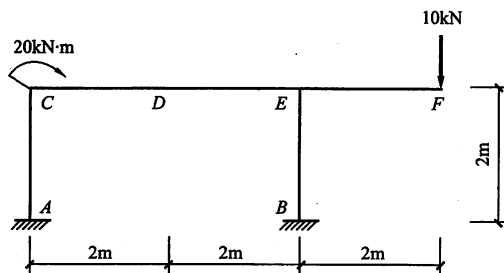
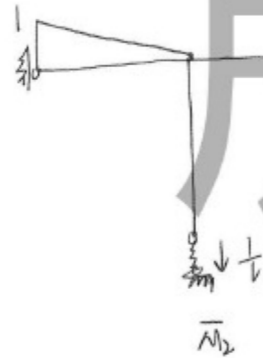
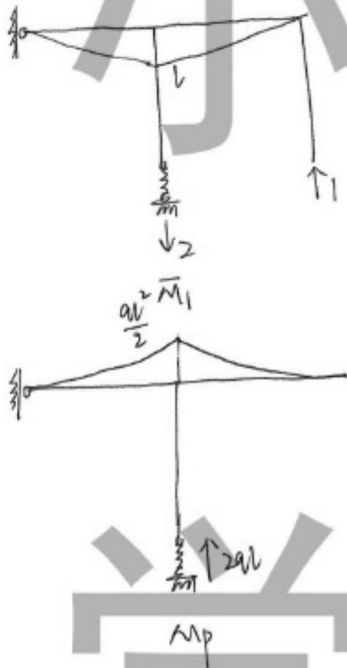


图 5 - 162

5-31. $\delta_{11}X_1 + \Delta_{1P} = -\frac{X_1}{k}$

5-32.



$$\delta_{11} = \frac{1}{EI} \left[\frac{1}{2} \times l \times l \times \frac{2}{3} \times \frac{2}{3} \right] + \frac{2 \times l}{k}$$

$$\delta_{11} = \frac{2l^3}{3EI} + 4 \times \frac{l^3}{3EI} = \frac{2l^3}{EI}$$

$$\delta_{22} = \frac{1}{EI} \left[\frac{1}{2} \times l \times l + 1 \times \frac{2}{3} \right] + \frac{1 \times \frac{1}{k}}{k}$$

$$\delta_{22} = \frac{l}{3EI} + \frac{1}{l^2} \times \frac{l^3}{3EI} = \frac{2l}{3EI}$$

$$\delta_{12} = \delta_{21} = -\frac{1}{EI} \left[\frac{1}{2} \times l \times l \times \frac{1}{3} \right] + \frac{2 \times \frac{1}{k}}{k} = -\frac{l^2}{6EI} + \frac{2}{l} \times \frac{l^3}{3EI} = \frac{l^2}{2EI}$$

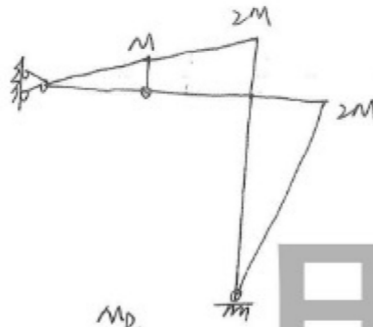
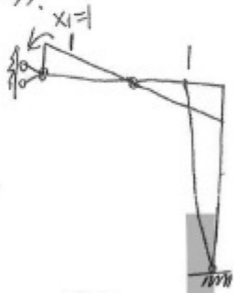
$$\Delta_{1P} = -\frac{1}{EI} \times \frac{1}{3} \times l \times \frac{ql^2}{2} \times l \times \frac{3}{4} \times 2 - \frac{2 \times 2ql}{k} = -\frac{ql^4}{4EI} - 4ql \times \frac{l^3}{3EI} = -\frac{14ql^4}{12EI}$$

$$\Delta_{2P} = \frac{1}{EI} \left[\frac{1}{3} \times l \times \frac{ql^2}{2} \times 1 \times \frac{1}{4} \right] - \frac{1 \times 2ql}{k} = \frac{ql^3}{24EI} - 2l \times \frac{l^3}{3EI} = -\frac{15ql^3}{24EI} = -\frac{5ql^3}{8EI}$$

$$\begin{cases} \delta_{11}X_1 + \delta_{12}X_2 + \Delta_{1P} = -\frac{X_1}{k} \\ \delta_{21}X_1 + \delta_{22}X_2 + \Delta_{2P} = 0 \end{cases}$$



5-33.

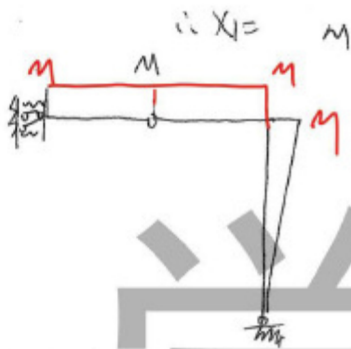


$$\delta_{11} = \frac{1}{EI} \left[\frac{1}{2} \times \frac{l}{2} \times 1 \times 1 + \frac{2}{3} \times 2 + \frac{1}{2} \times l \times 1 \times \frac{2}{3} \right] = \frac{1}{EI} \left[\frac{l}{3} + \frac{l}{3} \right] = \frac{2l}{3EI}$$

$$\Delta_{1P} = -\frac{1}{EI} \left[-\frac{1}{2} \times \frac{l}{2} \times 1 \times M \times \frac{1}{3} + \frac{1}{2} \times \frac{l}{2} \times 1 \times 2M \times \frac{5}{6} + \frac{1}{2} \times l \times 1 \times 2M \times \frac{2}{3} \right]$$

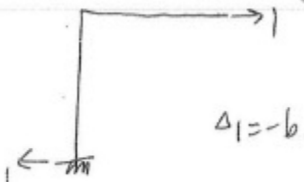
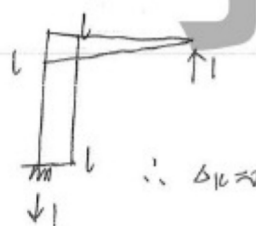
$$\Delta_{1P} = -\frac{1}{EI} \left[-\frac{Ml}{12} + \frac{5Ml}{12} + \frac{8Ml}{12} \right] = -\frac{Ml}{EI}$$

$$\delta_{11}x_1 + \Delta_{1P} = -\frac{x_1}{k_0} \quad \therefore \frac{2l}{3EI} x_1 - \frac{Ml}{EI} = -x_1 \cdot \frac{l}{3EI}$$

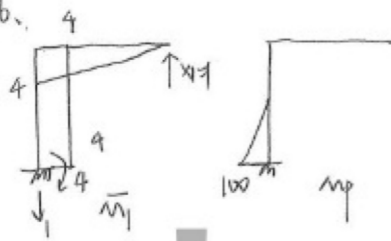


5-34. A

5-35.



5-36.



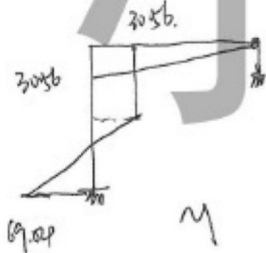
$$\delta_{11} = \frac{1}{EI} \left[\frac{1}{2} \times 4 \times 4 \times 4 \times \frac{2}{3} + 4 \times 4 \times 4 \right]$$

$$\delta_{11} = 20.32 \times 10^{-3} \text{ m/kN}$$

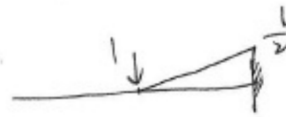
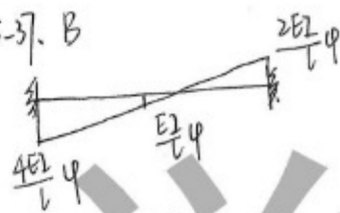
$$\delta_{1P} = -95.24 \times 10^{-3} \text{ m}$$

$$1 \times \delta_{11} + 1 \times 20 \text{ mm} + 4 \text{ mm} \times 0.0 \approx 0 \quad \therefore \delta_{11} = -0.06 \text{ m}$$

$$\delta_{11} X_1 + \delta_{1P} + \delta_{11} = 0 \quad \therefore X_1 = 7.69 \text{ kN}$$



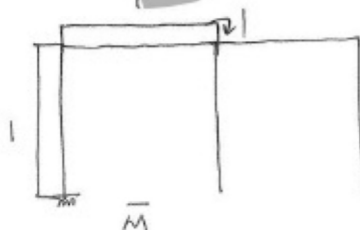
5-37. B



$$\Delta \phi_1 = \frac{1}{EI} \left[\frac{1}{2} \times \frac{1}{2} \times \frac{2EI}{l} \phi \times \frac{1}{2} \times \frac{2}{3} - \frac{1}{2} \times \frac{1}{2} \times \frac{EI}{l} \phi \times \frac{1}{2} \times \frac{1}{3} \right]$$

$$\Delta \phi_1 = \frac{1}{EI} \left[\frac{EI \phi l}{6} - \frac{EI \phi l}{24} \right] = \frac{\phi l}{8}$$

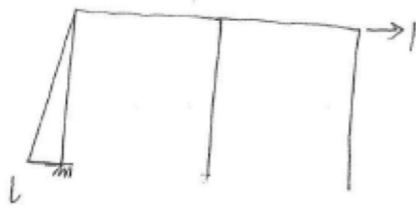
5-38. 作取一个单元体系图



$$\phi_c = \frac{100 \text{ kN}}{EI} \left[\frac{1}{2} \times 1 \times 7.29 \times 1 - \frac{1}{2} \times 1 \times 3.65 \times 1 + \frac{1}{2} \times 1 \times (7.29 + 11.98) \times 1 \right] - \frac{1}{EI} \times \frac{1}{2} \times 6 \times \frac{1}{4} \times 1$$

$$\phi_c = - \frac{1.045 \times 10^2 \text{ kN}^2}{EI}$$





$$\Delta H_E = \frac{0.01 Fl^3}{EI} \left[\frac{1}{2} \times 6 \times 7.29 \times 6 \times \frac{1}{3} - \frac{1}{2} \times 6 \times 3.65 \times 6 \times \frac{2}{3} \right]$$

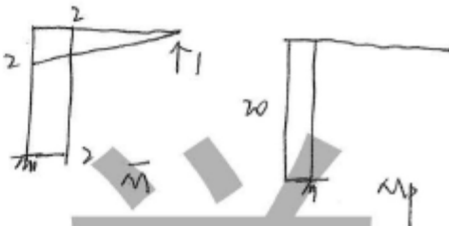
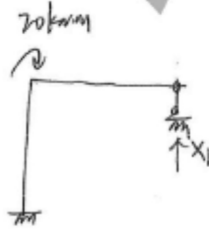
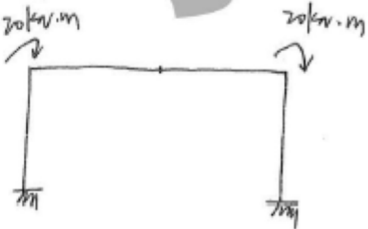
$$\Delta H_E = - \frac{1.67 \times 10^{-5} Fl^3}{EI}$$

5-39.



$$\Delta = \frac{1}{EI} \left[\frac{1}{2} \times 6 \times 6 \times \frac{EI \theta}{l} \right] = \frac{l \theta}{2}$$

5-40. (1)

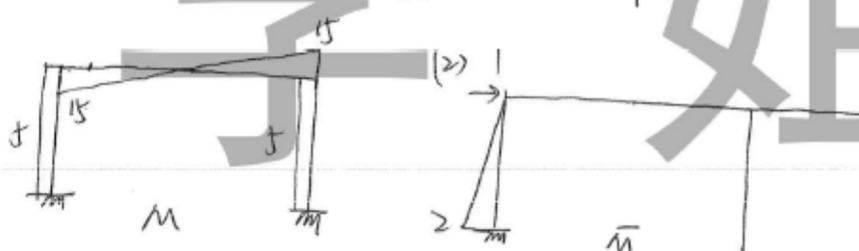


$$\delta_{11} = \frac{1}{EI} \left[\frac{1}{2} \times 2 \times 2 \times 2 \times \frac{2}{3} + 2 \times 2 \times 2 \right]$$

$$\delta_{11} = \frac{1}{EI} \left[\frac{8}{3} + \frac{24}{3} \right] = \frac{32}{3EI}$$

$$\delta_{1P} = - \frac{1}{EI} [2 \times 2 \times 20] = - \frac{80}{EI}$$

$$\delta_{11} X_1 + \delta_{1P} = 0 \quad \therefore X_1 = 7.5$$



$$\Delta u_x = \frac{1}{EI} \left[\frac{1}{2} \times 2 \times 2 \times 15 \right] = \frac{10}{EI}$$



据结构与荷载的特点，选用最简便的方法作刚架的弯矩图；（2）求 C 点的水平位移。（武汉大学 2011）

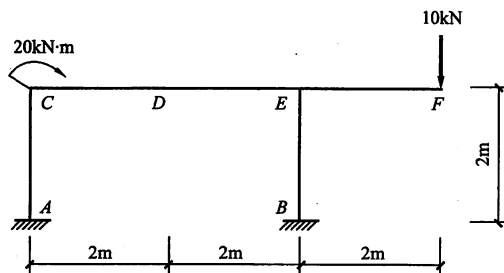


图 5 - 162